

Trading Strategies — Living Document

A running catalogue of every signal/strategy we've considered, built, tested, or shelved — with **status and evidence**, so we refine over time instead of re-litigating. Update the changelog at the bottom on every material change.

Golden rule: nothing earns risk until it survives **point-in-time features + friction-aware, regime-split backtesting**. The broad-sentiment edge looked great and died exactly this way — see `backtest/engine.py` .

Status legend

	Meaning
✔ Validated	Survived friction-aware, regime-split backtest (still preliminary — see caveats)
● Marginal	Real effect, but fragile net of costs / not sign-stable
✖ Failed	Tested and does not work (kept here so we don't retry it)
🧪 Untested	Built and wired, awaiting data to evaluate
💡 Idea	Backlog — not yet built
🔄 In progress	Data/build task underway

Progress at a glance

Last updated: 2026-06-02. Update this section whenever a status, metric, or next-step changes.

Strategy scoreboard

Strategy	Class	Status	Headline result	Next action
<code>retail_fear</code> (capitulation / bear-extreme / pessimism-fade)	Macro overlay	✅ Validated (re-confirmed 2016-26)	excess vs buy-&-hold across a risk-on basket. AUDJPY (core edge) on the extended 2016-26 series: +0.48%/10d VIX-gated, t(exc)=2.93, +ve in 9/10 years (2018 the lone falling-knife year, -1.05%). Gold +0.6 / SP500 +0.2 robust; Silver +2.2 lumpy; VIX gate ~doubles it	live via MT5 <code>RedditMacro_EA</code> ; vol-target sizing
<code>rrai_momentum_long</code> , <code>froth_high_long</code>	Macro overlay	❌ No edge	positive net but ~0 excess vs buy-&-hold (just bull drift)	dropped
<code>rrai_euphoria_short</code> + all macro shorts	Macro overlay	❌ Failed	-1.5 to -3pp excess (shorting the bull)	dropped
<code>dump_fade_DELETION_short</code>	Equity	❌ FAILS at full coverage (was a coverage-bias artifact)	After profiling all 487 pump-authors (25%→100%) the edge flips negative : gone $\geq 0.20 \rightarrow -19.3\%/10d$ (n=87), dominated by squeeze tails (IXHL -625%); <code>PIT young_frac</code> short is -39% to -87% (young pumps rip <i>hardest</i>). The earlier +14.3% (n=36 @ 25% cov) simply hadn't profiled the squeezers. Win rate still 64% but uncapped stock-short tails kill it.	dead as a stock short ; only conceivable via puts (caps the -625% tail) — <code>distribution_short</code> is the better-behaved version (price-rolling-over filter dodges live squeezes)
<code>distribution_short</code> (concentrated pump + sentiment-price divergence)	Equity	🟡 Promising — PIT-tradeable	concentrated pump (per_author ≥ 2 , m ≥ 10) + sentiment ≥ 0.4 + 5d price already rolling over $\rightarrow +7.2\%/10d$ net short, t=2.70; date-clustered t=2.52 (47 indep. dates), dose-response in sentiment, +ve all 3 sub-periods, survives 2x costs ; control (same pumps, no divergence) <i>loses</i> -4.1%	OOS-collect forward; tighten <code>per_author</code> (some ETF/large-cap leak in); express via puts; promote on more episodes
<code>fade_organic_short</code>	Equity	🟡 Marginal	+0.83%/5d but P3 flips	retry via puts + VIX/vol gate
<code>pump_long</code> (ride)	Equity	❌ Failed	+3% but regime-flips, t=0.75	— drop
<code>dump_fade_short</code> (naive)	Equity	❌ Failed	-9.9% (squeezed)	superseded by deletion-gated
<code>organic_long</code>	Equity	❌ Failed	-4.2%, sign-stable loser	inverse $\rightarrow$ <code>fade_organic_short</code>

Build & data progress

Component	Status	Detail
Feature store (<code>feature_daily</code> / <code>aggregate_daily</code>)	✓ Done	~77k ticker-day rows, point-in-time
Friction-aware backtester (<code>backtest/engine.py</code>)	✓ Done	costs + regime split + no-lookahead
<code>/signals</code> feed (multi-strategy, json mt5)	✓ Done	strategy-tagged; live, currently flat
Macro strategy R&D (<code>backtest/macro_research.py</code>)	✓ Done	only <code>retail_fear</code> beats buy-&-hold; momentum/froth no edge
MT5 framework (<code>SignalClient.mqh</code> + <code>RedditMacro_EA.mq5</code>)	✓ Done	strategy-selectable thin client; demo-only
Author profiling coverage	📊 24% overall / 100% of pump-authors	46,637 profiled, 6,153 gone; sweep now skips crypto-only + front-loads ticker-mentioners. Targeted-profiled all 487 equity pump-authors.
Deletion-gated short	✗ Failed at full coverage	The +14.3% was a coverage artifact (25% sample missed squeezers). At 100% pump-author coverage: gone $\geq$ 0.20 – 19.3%/10d (IXHL –625% tail); young_frac short –39%→–87%. Stock-short dead; puts-only if at all.
Short-interest / squeeze data (FINRA Reg SHO)	✓ Done	<code>scraper/finra.py</code> : 475k rows / 5,693 tickers; z-score screen (abs level is MM noise, median 0.49)
Independent fear-confirmation gate (Wikipedia)	🟡 → 🟢 Strengthened on 2016-26	Extended series (525 caps, <code>backtest/fear_gate.py</code>): fear-only excess SPY +0.57 (day t=1.64, episode t=2.77) / QQQ +0.80 (day t=1.91, episode t=2.37) — effect <i>grew</i> with the 2018 episodes & SPY/QQQ now agree ⇒ likely real (day-level still marginal). The LIVE trend+fear still ≈0 (SPY +0.10/QQQ +0.11) → trend filter should be dropped, use fear-only . AUDJPY stays ungated (fear <i>hurts</i> it, +0.21→+0.09).
Live screens in <code>/signals</code> (<code>tradeable:false</code>)	✓ Done	<code>smallcap_pump_fade</code> · <code>fresh_pump_alert</code> · <code>coordination_flag</code> · <code>biotech_catalyst</code> · <code>attention_fade</code> · <code>calendar_overlay</code>
Idea backlog	💡 7 ideas	squeeze · options-flow · hype-cycle · sentiment-extremes · rotation · novelty · coordination
Equity/options paper client (Alpaca)	💡 Planned	not started

Methodology (non-negotiables)

1. **No lookahead** — features at date D use only data $\leq$ D; entries at the first close *after* D.
2. **Frictions** — round-trip spread (small-caps wide), short borrow, halts, capacity.
3. **Regime split** — net return per sub-period; sign-stability is the bar, not a big average.
4. **Shorts via PUTS**, not naked stock — penny shorts have squeeze/borrow/halt tails.
5. **Overlapping-window caveat** — daily signals with N-day forward returns inflate t-stats.
6. **Benchmark macro vs buy-and-hold** — in a bull market "long anything" looks like alpha; a macro long must beat *unconditional drift* (excess), not zero. (Cost us 5 false "survivors".)

Asset-class reality (what this data can/can't trade)

- **Small/micro-cap & meme equities** — primary signal (retail flow moves price).
- **Single-name options** — primary *expression* (convex longs, defined-risk shorts).
- **Risk-FX (AUD/NZD complex)** — the **primary macro expression**: retail capitulation → contrarian bounce, validated across the cross-section (AUDJPY/NZDJPY/AUDUSD/AUDCHF, t 2.7-3.2). The cleanest tradeable read of the risk-appetite factor RRAI

measures.

- **Index / metals** — *macro overlay* (Gold robust; equities the weak expression, $t \approx 1$).
- **Haven/non-risk FX (USDJPY, CHFJPY, EURUSD) / "Reddit sentiment on SPY as direction"** — no edge; don't build it (they're the controls that confirm the factor).

1. Macro overlay — Retail Risk-Appetite Index (RRAI)

Aggregate retail positioning as a **contrarian** factor at extremes. Built posts-only + ratios (coverage-stationary); 90-day trailing percentile. `analytics/features.py::aggregate_daily`.

Macro R&D verdict (`backtest/macro_research.py` , **excess over buy-and-hold).** A bull market made *every* macro long look like a winner and *every* short a loser — until benchmarked against unconditional drift. Net of that, only the **contrarian retail-fear** family beat buy-and-hold. RRAI-momentum and froth had ~ 0 excess (pure drift); all shorts were -1.5 to -3 pp (fighting the trend). **One real macro edge, three formulations of it.**

✓ `retail_fear` — buy risk when retail capitulates (*served live*)

- **Thesis:** retail max-bearish (RRAI pct low / bear-share extreme) + genuine fear (VIX high) → contrarian bounce. Captured 3 ways (capitulation, bear-extreme, pessimism-fade) — all positive-excess, mutually corroborating.
- **Spec (shipped):** long when `rrai_pct` ≤ 0.15 **AND** `VIX` ≥ 18 ; hold ~ 10 d; size by extremity; **fear-side only**.
- **Evidence (12mo, net, 10d): excess over buy-&-hold** SPY +0.17pp / QQQ +0.23pp / **AUDJPY +0.47pp** (win 88%); `bear_extreme` +0.24/+0.32/+0.36pp corroborates. **VIX gate $\sim 4\times$ 'd it** (SPY +1.56%/10d in fear vs +0.42% calm). Modest **timing tilt**, not standalone alpha (buy-&-hold already won 66–75%).
- **Instrument scan (17 MT5 instruments, 10d excess vs buy-&-hold):** the signal **generalises across the risk-on basket** — Silver +2.22pp (lumpy, one big rally → size small), Gold +0.59pp (robust, +ve every regime), AUDJPY +0.47pp (steadiest, win 88%), SP500 +0.17pp. Oil (+1.82pp but one-episode) and crypto (BTC net-negative) **dropped**. Generalisation across assets is itself evidence it's a real risk-appetite signal.
- **★ Cross-sectional FX validation (2026-06-03, 2016-26, VIX-gated) — proves it's a RISK-ON-CURRENCY factor, not an AUDJPY fluke.** Ran capitulation-long across the yen-cross + risk-pair complex. The edge travels with the **risk-on currency (AUD/NZD)**, independent of the funding leg — and the **controls fail**, which is the real proof:

pair	character	excess	t(exc)
NZDJPY	risk-on JPY cross	+0.43%	+3.18
AUDJPY	risk-on JPY cross	+0.48%	+2.93
AUDCHF	risk vs CHF-haven	+0.41%	+2.83
AUDUSD	risk pair (no JPY)	+0.43%	+2.72
EURJPY	risk-ish	+0.25%	+2.70
NZDUSD	risk pair	+0.35%	+2.33
Nikkei ^N225	high-beta JP equity	+0.62%	+1.93
USDJPY	JPY cross, USD~haven	+0.12%	+1.02 (flat)
CHFJPY	CONTROL haven/haven	+0.13%	+1.37 (flat)
EURUSD	CONTROL non-risk	+0.14%	+1.44 (flat)

Read: every AUD/NZD pair is significant (t 2.7–3.2) regardless of funding leg; the three pairs with *no* risk-on leg (USDJPY/CHFJPY/EURUSD) are the three weakest. **It's not a JPY thing** — JPY is just a clean funding leg; the Nikkei works because it's high-beta *risk*, not because it's Japanese. Equities stay the weak expression (SPY $t=0.90$, QQQ $t=1.13$). The clean cross-section (right pairs fire, controls don't) is strong evidence of a genuine factor → **trade the basket, not one pair**. - ● **Independent fear-**

confirmation gate (Wikipedia, LIVE — but now DOUBTFUL): the **equity** legs (US500/USTEC) additionally require an *independent* fear spike — $\text{fear_z} \geq 0.5$, the z-score of Wikipedia fear-page attention (Stock_market_crash / Recession / Bear_market views) vs the trailing ~30d — **AND** a 200-DMA uptrend. Original rationale: SPY capitulation *alone* doesn't beat buy-and-hold (master backtest excess -0.05); an *ad-hoc* run reported **+0.36pp/10d when fear_z confirms vs ~-0.01 ungated**, and fear-attention is **1.32x higher on capitulation days** (DATA_PLAN.md). Wired live in analytics/signals.py (_fear_z , FEAR_MIN ; fear_z in /signals). -  **MT5 Strategy-Tester A/B (US500, D1, 2020-2026, long-only, real spread/swap; mql5/RedditMacro_US500_*.ini , dates via python main.py --export-signals)**. This tests the gate as an **absolute long-only rule** (NOT excess-vs-buy-&-hold):

```
| variant | signals | trades | win% | net $ | PF | maxDD% |
|-----|-----|-----|-----|-----|-----|-----|
| baseline (all caps) | 335 | 100 | 57.0 | **+12** | 1.00 | 13.6 |
| 'fear_z>=0.5' only | 95 | 45 | 53.3 | **~1045** | 0.68 | 13.0 |
| trend up (200-DMA) only | 223 | 71 | 56.3 | **~21** | 1.01 | 9.0 |
| trend **AND** fear (the live gate) | 54 | 27 | 55.6 | **~372** | 0.73 | 7.1 |

**Result: the 'fear_z' gate does NOT help - it hurts.** Adding fear makes US500 *worse* both alone (-$1045)
```

-  **Committed EXCESS study (backtest/fear_gate.py , python -m backtest.fear_gate) — this settles it.** The frame MT5 can't measure: mean EXCESS over buy-&-hold (the +0.36pp's own metric). Same point-in-time date-sets as the MT5 run (shared capitulation_sets()). 10d, net of costs:

leg	variant	n	excess%	win%	t(exc)
SPY	baseline	332	-0.05	61	-0.21
SPY	fear	94	+0.32	68	+0.64
SPY	trend	220	-0.03	60	-0.16
SPY	trend+fear (<i>live</i>)	53	+0.09	64	+0.19
QQQ	baseline	332	+0.06	62	+0.20
QQQ	fear	94	+0.41	65	+0.71
QQQ	trend+fear (<i>live</i>)	53	-0.20	60	-0.33
AUDJPY (<i>ungated</i>)	baseline	333	+0.26	68	+2.04

Verdict (three honest parts): (1) **The +0.36pp REPRODUCES as a point estimate** — fear-only SPY +0.32 / QQQ +0.41 excess — so it was *not* fabricated; the fear gate does lift equity-capitulation excess. (2) **But it is NOT statistically significant** (excess $t \approx 0.6$, $n \approx 94$) — a thin, unproven edge. (3) **The LIVE trend+fear combo is the wrong way to apply it:** the trend filter *removes the buy-fear-in-a-downtrend bounces that carry the excess*, diluting SPY (+0.32 → +0.09) and flipping QQQ negative (+0.41 → **-0.20**). - **Reconciliation (why MT5 said "loss" and this says "small win"):** MT5 sums *absolute* P&L and holds one position at a time, so a few COVID-cluster trades dominate the dollar total (-\$1045); the excess frame means-averages independent trades, where fear is mildly +ve. Both are correct for their question — for *signal research* the excess frame is the right one, so the earlier "fear-gate is a drag" was a frame artifact. - **The genuine tension:** fear → better *timing/excess* but worse *drawdown* (COVID knives); trend → better *drawdown* but worse *excess*. The live **trend AND fear** gets the worst of both for excess. - **Recommendation:** keep the gate  (positive but insignificant). If we ever optimise the equity legs for excess, use **fear-only, not trend+fear**; AUDJPY (the one significant edge, +0.26 $t=2.04$) stays **ungated** — confirmed correct. (User chose to leave the live gate unchanged for now; this is logged for when we revisit.) -  **Power / effective sample (day vs episode — the divergence IS the verdict):** the 94 fear-days are *not* 94 independent obs — overlapping 10d windows + episode clusters. Treated as **days**, fear-only is $t=0.64$ (SPY)/0.71 (QQQ); collapsed to **independent episodes** (~40, one vote each) it's **$t=2.28$ (SPY)/1.85 (QQQ)**. The gap is driven almost entirely by the **COVID-2020 cluster** (~10+ losing fear-days → one episode vote), so the effect is *positive across most episodes but fragile* — and SPY (2.28) vs QQQ (1.85) disagreeing on near-identical instruments over the same episodes says we're at the **noise floor**, not a stable edge. **Data to settle it:** day-level needs ~9x more obs for $p < 0.05$ (~19x for 80% power) — impractical (~40+ yrs at ~15-20 fear-days/yr); the real lever is **independent fear episodes**, and the cheapest source is *backward* (Wikipedia fear data already reaches 2015) — **extending the RRAI/capitulation**

series to 2016-2017 adds the 2018 Volmageddon + Q4-2018 episodes (price coverage starts 2016-05). -  **DONE — RRAI extended to 2016-06 (525 caps, +190)**. Re-ran the study on the longer series, and the fear-gate **strengthened** (effect grew out-of-sample — the hallmark of a real edge):

leg	fear excess (94d → 141d)	day t (→)	episode t (→)
SPY	+0.32 → +0.57	0.64 → 1.64	2.28 → 2.77
QQQ	+0.41 → +0.80	0.71 → 1.91	1.85 → 2.37

Both equity legs are now episode-level significant (t≈2.4-2.8) AND agree — the earlier SPY≠QQQ noise-floor red flag is gone. Day-level is still only marginal (1.6-1.9), so not fully nailed, but this is a real upgrade from "fragile/unproven." The trend filter conclusion **hardens**: `trend+fear` (live) is still ≈0 excess (SPY +0.10/QQQ +0.11) ⇒ **drop trend , use fear-only** on the equity legs. AUDJPY confirmed best ungated (baseline +0.21 t=2.00; fear-gating *hurts* → +0.09). - **Expression: a diversified risk-on basket** — now **AUDJPY + NZDJPY + AUDUSD** (the AUD/NZD risk-pair complex) + Gold (XAUUSD) + US500/USTEC + small Silver (XAGUSD). Served via `/signals` (tag `retail_fear`); traded by `mql5/RedditMacro_EA.mq5` (demo). FX/metal legs ungated; **equity legs gated by `fear_z` only** (the 200-DMA trend filter was dropped 2026-06-03 — it killed the excess; see the fear-gate row). - **Caveats**: single ~12mo bull regime; **46 cap-days = only ~13 distinct fear episodes** (clustered → effective n small); overlapping windows; edge small in absolute terms. - **Next**: leverage-language enrichment (calls/puts, margin/YOLO); longer history (data-limited); vol-target sizing; combine capitulation+bear-extreme into one composite trigger.

 Macro candidates that failed (kept so we don't retry)

- `rrai_euphoria_short` , `bull_extreme_short` , `froth_high_short` , all momentum-shorts: **-1.5 to -3pp excess** — shorting a bull market.
- `rrai_up_momentum_long` , `froth_high_long` : **~0 excess** — just drift, no signal.

2. Equity cross-sectional (per ticker × day)

From `analytics/features.py::feature_daily` (velocity, accel, breadth, concentration/HHI, sentiment trajectory, author young/gone/suspended mix, novelty).

 `pump_long` — **ride concentrated promotion**

- **Thesis**: concentrated mention spike (few authors, high volume) pops short-term.
- **Result**: net +3%/5d but **sign-flips** (P1 +15.8 → P3 -4.6), win 36%, t=0.75. The crude "+6.45%" was a no-lookahead/no-cost/single-regime artifact. **Not robust**.

 `dump_fade_short` — **short a decelerating pump (naive)**

- **Result**: net **-9.9%** (P1 -29%). Shorting pumps on deceleration gets squeezed/run over. Wrong trigger; brutal without defined risk.

 `organic_long` — **buy broad attention spikes**

- **Result**: sign-stable **-4.2%** net (t=-4.25, n=321). Buying broad/news-driven attention tops is a *consistent loser* → the inverse is the candidate (below).

 `fade_organic_short` — **fade broad attention spikes**

- **Thesis**: broad attention spikes mean-revert down (inverse of the robust `organic_long` loss).
- **Result**: net +0.83%, but **flips in P3** (win 46%) — the reversion is real but shorting costs + recent weakening eat it. **Marginal**. Revisit via **puts** (cheaper than borrow) and with a VIX/vol gate.

✗ `dump_fade_DELETION_short` — short a pump whose promoters vanished (*THE thesis — DIED at full coverage*)

⚠ **REVERSAL (2026-06-03)**. This was "forensically validated" at +14.3%/10d (n=30-36) — but that ran at only ~25% pump-author coverage. We then **targeted-profiled all 487 equity pump-authors (→100%)** and re-tested: the edge **flips to -19.3%/10d (n=87)**, and the PIT `young_frac` short is -39% to -87%. Cause: **squeeze tails** — a single ticker (IXHL) squeezed **-625%/-404%/-377%** in July-2025; the partial sample simply hadn't profiled its authors, so those events were invisible. Lesson logged: **small-n + partial-coverage results are dangerous**; shorting live manipulated micro-caps has *uncapped* left-tail risk. Win rate is still ~64% (most pumps fade), so the thesis isn't *wrong* — but it's only expressible via **puts** (defined risk caps the -625%). The price-rolling-over filter is what makes `distribution_short` survive where this dies. - **Thesis (original)**: concentrated pump + promoters now **deleted/suspended** → manufactured → fade the collapse. The original A2 idea, and the *reason* we track account lifecycle. - **Result (2026-06-02, 20% author coverage, h=10, market-neutral, 75bps + 8bps/day borrow)**: | gate | n | net short | win | t | regimes | |---|---|---|---|---|---| | `gone_frac`≥0.10 | 33 | +10.0% | 81.8% | 1.51 | -1.2 / +19.9 / +11.4 | | `gone_frac`≥0.20 | 30 | +14.3% | 83.3% | 2.57 | +11.4 / +27.7 / +3.9 | | **control**: pumps w/ promoters *still here* (`gone`=0) | 202 | -10.8% | 36.1% | -2.22 | -30.7 / -0.6 / -1.1 | - The gate **flips the sign of the trade** (~+25pp spread). Un-gated concentrated pumps are *un-shortable* (they continue / squeeze); only the ones whose promoters vanish collapse. The deletion is doing the work, not pump-selection. Positive across all 3 regimes at `gone`≥0.20. - ⚠ **Not yet point-in-time tradeable**. `gone_frac` uses *hindsight*: Reddit exposes no deletion timestamp, so "gone" means "gone as of our latest sweep" — unknowable at the pump date. The PIT-available leading proxy is `young_frac` (account age at post time), but it's currently too sparse to conclude (n=3-7, and the few events lean the wrong way). - **Path to tradeable**: (1) **finish the author sweep** (20%→40%+) to power the `young_frac` PIT test; (2) failing that, a **live deletion-monitor** rule — `author_status_revalidate` already detects deletions forward; short on deletion-*confirmation* within K days of a fresh pump, capturing the back half of the collapse. Express via **puts** (defined risk vs squeeze tails; many are HTB/no-borrow).

● `distribution_short` — concentrated pump + sentiment-price divergence (*NEW, point-in-time tradeable*)

- **Thesis**: the "bagholder distribution" moment — a *manufactured* pump (concentrated chatter) where **price has started rolling over but the crowd is still euphoric**. Insiders distribute into the retail bid; sentiment lags price. Every input (`per_author` , `mentions` , `sentiment` , trailing price return) is known at signal time → **unlike the deletion gate, this is actually tradeable live**. (`backtest/microstructure.py`)
- **Rule**: `per_author`≥2 AND `mentions`≥10 AND `sentiment`≥0.4 AND `5d-trailing-return`<0 → SHORT. h=10, market-neutral, 75bps + 8bps/day borrow.
- **Result (2026-06-02)**: +7.2%/10d net, t=2.70, win 52%, n=60 across 39 distinct tickers.
- **Date-clustered test** (1 obs/date, kills intra-day correlation): +7.6%, t=2.52, 47 independent dates — *not* a risk-off-day artifact (max 3 events on any date, spread over 12 months).
- **Dose-response** in sentiment (`s`≥0.3/0.4/0.5 → +3.9/+7.2/+11.1%) and **monotone in horizon** (h=5/10/20 → +5.6/+7.2/+9.5%, t rising to 2.92) — the hallmarks of a real effect, not a fit.
- **Survives 2x costs** (150bps spread + 16bps/day borrow → still +4.9%/10d).
- **Control**: same concentrated pumps *without* the divergence (price up) → -4.1% — the divergence is doing the work. Positive in all 3 sub-periods.
- **Why it works where the others don't**: the gate is *precise* — it only fires inside the manufactured subset. The broad high-attention universe **drifts flat-to-up** (gross ≈ +0.04%), so generic pump-shortening just pays costs (see negatives below); the edge requires isolating the distributing pumps, which both this and the deletion gate do.
- **Caveats / next**: n still modest (47 indep. dates); single ~12-mo window (sub-periods, not true bull/bear regimes); small-cap-heavy so no multi-regime price history; loose `per_author`≥2 lets some ETFs/large-caps (QQQ/IWM) leak in — tighten to ≥3 and/or split small-cap vs index. **Collect forward OOS** (the clock starts now); express via **puts**. Not yet promoted to the live `/signals` feed.

✗ **Negatives from the same study (recorded so we don't re-run them)**

- **Broad-universe microstructure timing does NOT survive costs**. Recruitment-exhaustion, mention-deceleration, and sentiment-price divergence applied to the *whole* high-attention set (`mentions`_z≥1) are all **net-negative** — the universe drifts

flat-to-up, so shorting it generically just bleeds the ~2.3% friction. Precision (concentration filter) is mandatory.

- **Recruitment-exhaustion alone = null.** New-author inflow rolling over is directionally supportive *within* concentrated pumps (+1.8% net, but $t=0.33$) and adds to the stack, but is not a standalone edge at current coverage. The new-author feature is built and kept.
- **Up-leg longs both fail.** (a) "Ride while recruiting + accelerating" (concentrated) netted +1.6% but win 27%, regimes [+21/-7/-9] — one big winner, otherwise negative. (b) "Attention-leads-price coil" (broad: recruitment surging while price still flat) was **net -1.4 to -1.9%** across horizons, sign-stable loser. Neither is a tradeable long; dropped.

3. Idea backlog (💡 not yet built)

- **Squeeze setup (long):** Reddit squeeze chatter (Shortsqueeze/SqueezePlays) + **short pressure** → squeezable. *Partially built:* `scraper/finra.py` + `squeeze_screen()` flag tickers with an anomalous **z-score jump** in daily short-ratio vs their own baseline (HTZ/GME currently surface). *Still needs* true SI%/float/days-to-cover (FINRA **bi-monthly short-interest** report) + borrow-fee for the level; the daily file's absolute level is market-maker noise. *Not yet backtested.*
- **Options-flow signal:** parse contract-level chatter (unusual_whales/options) → retail-gamma/positioning proxy. *Needs contract parsing.*
- **Hype-cycle lifecycle:** model attention as nascent→acceleration→peak→fade; long the *organic* acceleration, fade the *concentrated* peak. Breadth-vs-concentration is the discriminator.
- **Conditioned sentiment extremes (single-name):** linear sentiment is dead, but *extreme concentrated euphoria at a price high* (short) / *capitulation after a sharp drop* (bounce) may be non-linear edges.
- **Cross-sectional attention rotation:** rank meme universe by abnormal velocity z-score; long top-decile / short bottom-decile, beta-hedged with SPY; baskets from co-mention clusters.
- **Novelty / first-mention alpha:** the "discovery" moment (silent → sudden organic burst).
- **Coordination graph (short):** rigorous version of the deletion short — cluster density + temporal bursts + duplicate text + new-account% + gone%.

4. Data gaps blocking strategies

Gap	Unblocks
Intraday prices (have EOD only)	timing pump/dump entries/exits (they move intraday)
Short interest / float / borrow fee	squeeze <i>level</i> (FINRA daily short-volume now in via <code>scraper/finra.py</code> , but it's MM noise at the level; need bi-monthly SI%/float + borrow)
Options chains / IV	options expression; options-flow signal
Longer history (data-limited to ~12mo)	true multi-regime validation of the RRAI overlay
Author coverage (20% → target ~40%+)	a <i>point-in-time tradeable</i> deletion short — thesis already forensically validated; PIT <code>young_frac</code> proxy needs more coverage

5. Execution stack

- `/signals feed` (`analytics/signals.py`, GET `/signals json|mt5`) — live, serves the capitulation overlay. ✅
 - **MT5 EA** (`mq15/RedditRRAI_EA.mq5`) — thin client for the index/FX overlay; demo-only. ✅
 - **Alpaca / IBKR paper client** — for the equity/options legs. 💡 planned.
-

★ MULTI-REGIME VALIDATION (2020-2026, incl. COVID crash + 2022 bear)

The deep-history backfill let us test across regimes — the result that actually moves confidence: - **AUDJPY capitulation-long is REGIME-ROBUST** — positive in EVERY year 2020-2026, including the 2022 bear (+0.7%, win 60%), win rates 60-93%. Graduates from preliminary → **validated**. This is the core tradeable edge. - **Equity (SPY) capitulation-long FAILS in a sustained bear** (2022: -0.7%/10d, win 43%) — falling-knife problem → **must gate with the 200-DMA trend filter** (don't buy fear in a downtrend). Confirmed needed. - **Euphoria-short only works in the bear** (2022 +1.3%; loses every bull year) → shorts are **regime-conditional**, switch on only in confirmed downtrends. (Vindicates not building shorts earlier.) - Caveats: small n/year (18-60), overlapping windows, historical RRAI is a WSB-heavy sampled backfill.

Session findings — sector / calendar / latency (2026-06-01)

- **Sector segmentation rescues the pump signal** (`backtest/sectors.py`). The broad pump was a wash because it *mixed opposites*: **small-cap concentrated pumps FADE** (−14.6%/10d excess, win 22%, $t = -3.93$ — pump-and-dumps reliably reverse → a short/avoid), while **biotech pumps are a fat-tailed lottery** (+66% mean, win 48%, $t = 2.23$ — a few catalyst moonshots). Thematic universes (ai/quantum/space/nuclear/meme/frontier) had too few concentrated-pump events to judge.
- **Pump detection is TOO SLOW** (`feature_daily`). Median `novelty_days` at flag = **152**; only **1%** of pumps flagged within 3 days. Root cause: the `min_total ≥ 20` universe filter + trailing z-windows mean a ticker must accumulate ~20 mentions before it's even eligible → we catch *established-ticker spikes*, not fresh pump-and-dumps. Only **33 fresh day-0 bursts** exist in the data. **Fix**: a fresh-burst fast-path that scans raw mentions (ticker's first day with a burst), bypassing the 20-mention gate — for LIVE catching, not just post-hoc.
- **Calendar effects (10y, SPY/QQQ/Gold; absent in AUDJPY-FX)**: turn-of-month is real (SPY +0.49 vs +0.20%/5d, QQQ +0.58 vs +0.35, Gold +0.38 vs +0.25); September weak (−0.2%); OpEx week underperforms. Generic equity-flow seasonality → useful as *overlays/filters*, not Reddit alpha.
- **AU sentiment track** building (`au_aggregate_daily` , post-level) to test AU-sentiment → AUDJPY.

★ MASTER BACKTEST vs BUY-AND-HOLD (`backtest/run_all.py` , full 2020-2026, 10d, net of costs)

Every signal judged on EXCESS over its instrument's unconditional drift (buy-and-hold-anytime). | Strategy | n | net% | B&H% | excess | win | verdict | |---|---|---|---|---|---| | retail_fear **AUDJPY** | 338 | +0.53 | +0.27 | **+0.25** | 69% | beats (modest, most robust) | | retail_fear SPY | 338 | +0.54 | +0.59 | −0.05 | 61% | **no edge** | | retail_fear Gold | 338 | +0.83 | +0.72 | +0.10 | 57% | marginal | | euphoria_short SPY | 351 | −1.26 | +0.59 | −1.84 | 27% | no edge | | smallcap_pump_fade | 87 | +9.07 | +0.59 | +8.48 | 64% | beats but single-regime + brutal costs | | biotech_catalyst | 91 | +22.2 | +0.59 | +21.6 | 36% | fat-tailed LOTTERY (low win) | | attention_fade | 252 | −0.73 | +0.59 | −1.31 | 40% | no edge |

Recalibration: with a proper buy-and-hold benchmark over the full period, the AUDJPY edge is a **modest +0.25pp/10d** (earlier raw "positive every year" overstated it - that was largely drift). SPY capitulation does NOT beat buy-and-hold. The big-excess equity signals are single-regime (smallcap) or low-win lotteries (biotech). Net: one modest-but-real macro edge, several promising- but-unvalidated screens. Do NOT size up on these as-is.

DATA WE NEED (to validate/strengthen — the binding constraints)

Data	Unblocks	Why we lack it
Intraday prices (we have EOD only)	pump entry/exit timing; fresh-pump fade execution (pumps move intraday)	not collected
Historical penny-stock prices + per-ticker features	multi-regime validation of <code>smallcap_pump_fade</code> & biotech (only 2025-26 exists)	delisted pennies → Yahoo gaps; per-ticker <code>feature_daily</code> only recent
Short interest / float / borrow fee	squeeze setups; realistic short cost/feasibility	FINRA daily <i>short-volume</i> in (<code>scraper/finra.py</code>); still need <code>SI%/float/borrow</code> (Ortex / FINRA bi-monthly SI)
Options chains / IV	express fades via PUTS (only sane way to short pennies); options-flow signal	not collected
More years of RRAI history	tighten the macro edge (effective $n \approx$ a few dozen fear episodes)	Reddit/arctic-shift depth ~2017+
Broader author coverage (~20% profiled)	a <i>PIT-tradeable</i> deletion/young-account short (thesis already validated forensically; <code>young_frac</code> proxy starved)	rate-limited Reddit /about sweep
Independent sentiment sources (StockTwits, news, Discord, crypto + AU subs)	independent episodes → real effective-sample growth without waiting calendar time	not integrated (AU track building)
Real transaction-cost data (per-instrument spreads/borrow)	trustworthy net returns (small-cap costs are modelled/optimistic)	not sourced

Update — complete-data confirmation (2026-06-01)

- US backfill finished: continuous 2020-2026 (2,344 days, 341 capitulation days). **Capitulation-long multi-regime result HOLDS on the complete series** — AUDJPY positive every year incl. 2022 bear (win 60-83%).
- `smallcap_pump_fade` is strong (+12.4% excess short, win 77%) but **only 2025-26 data exists** (n=87) — no historical penny-stock prices/features, so it **can't be multi-regime validated** → stays ADVISORY, NOT promoted to tradeable. (Single-regime; can't be fixed by backfill — penny price history doesn't exist.)

Crypto — coincident, NOT a leading indicator (tested 2026-06-03)

Added CryptoCurrency/Bitcoin/CryptoMoonShots/SatoshiStreetBets (~1yr; Bitcoin 61k posts) and tested whether crypto leads equities / risk-appetite. **It doesn't — it's coincident**: - **Price lead-lag (BTC↔SPY, 2,577d)**: same-day corr **+0.17 (t=8.8)** but every lead/lag $k \neq 0$ is noise ($|t| < 1.3$) — neither leads. - **Weekend test** (crypto's best shot — BTC trades Sat/Sun while equities are closed; 490 weekends): corr +0.05, **t=1.2** — no lead. - **Crypto sentiment → equity RRAI**: corr **peaks at k=0 (+0.20, t=3.9)** then decays (a lead would peak at $k > 0$). Barely predicts even BTC's *own* next-day return (+0.05). - **Combined crypto+equity RRAI capitulation (1yr overlap, n=55)**: does **NOT** help — for the validated AUDJPY edge it *dilutes* it (equity-only excess **+0.55%** > combined +0.45% > crypto-only -0.03%). Crypto sentiment is redundant with (and noisier than) the equity RRAI. - One weak whiff: crypto *euphoria* → mildly negative SPY 3d (t=-2.4, but overlapping windows → true $t \approx -1.4$) — *contrarian*, not leading. Watch, not an edge.

Takeaway: crypto's value is NOT macro timing (it's a redundant coincident risk-appetite read). It's worth keeping only as its own **pump/meme universe** (CryptoMoonShots/SatoshiStreetBets) for the `distribution_short` /pump signals. Do not fold it into the RRAI.

★ Cross-asset generalization map (2026-06-03) — it's a RISK-APPETITE / FEAR-REVERSAL factor

Capitulation-long ($\text{rrai} \leq 0.15$ & $\text{VIX} \geq 18$, 10d, excess vs buy-and-hold) tested across asset classes. The signal fires wherever the instrument is a clean **risk-on** or **fear-bid** play, and is flat everywhere else — the pattern of winners *and* losers is mechanistically coherent (= a real factor):

asset class	works?	evidence
Risk-on FX (AUD/NZD)	✅ strongest	AUDJPY $t=2.93$, NZDJPY $t=3.18$, AUDCHF 2.83, AUDUSD 2.72, NZDCHF 2.58, ZARJPY 2.32 (EM, noisier)
Safe-haven metals	✅ gold	Gold +0.43% $t=2.03$; Silver +0.76% $t=1.72$ (lumpy)
Equity indices (US/AUD/JP)	❌ weak	Nikkei +0.62 $t=1.93$ (best); S&P $t=0.90$, Nasdaq 1.13; ASX 200 $t=0.08$ (flat)
Growth commodities	❌ none	Copper $t=0.68$, Platinum 0.87, Oil -0.10
Controls (should be flat)	✅ flat	AUDNZD (risk/risk) $t=1.11$; USDJPY 1.02, CHFJPY 1.37, EURUSD 1.44

Punchlines: (1) the AUD *currency* carries the edge ($t=2.9$) but the AUD *equity index* (ASX) is flat ($t=0.08$) — it's an FX/metals phenomenon, not equities. (2) **AUDNZD flat** is the cleanest proof: two risk-on currencies cancel the factor. (3) It's ONE factor, so more instruments = diversification *within* it (diminishing), not new alpha; EM/growth-commodity legs add cost/tail-risk for little gain. **MT5 absolute check** (long-only, ungated, BaseLots=1.0): AUDJPY net +\$10.4k (PF 1.14) but AUDUSD ~flat in dollars (it *declined*, so +excess $\neq$ +absolute) and 60%+ DD → deployable version needs the VIX gate + vol-target sizing.

Changelog

- **2026-06-03 — Cross-asset generalization map + MT5 FX backtest.** Capitulation-long tested across FX/indices/commodities: it's a **risk-appetite/fear-reversal factor** — strong in risk-on FX (AUD/NZD, EM-ish), works in gold/silver, **flat in equity indices (incl. ASX $t=0.08$) and growth commodities (copper/oil)**; AUDNZD control flat. MT5 tester (D1, 2016-26, ungated): AUDJPY +\$10.4k/PF 1.14, AUDUSD ~flat-\$ (declined, +excess only), 60% DD → needs VIX gate + vol-target to deploy.
- **2026-06-03 — Cross-sectional FX validation → broadened the basket.** Tested capitulation-long across the yen-cross + risk-pair complex (2016-26, VIX-gated). The edge is a **risk-on-currency factor**: every AUD/NZD pair is significant (NZDJPY $t=3.18$, AUDJPY 2.93, AUDCHF 2.83, AUDUSD 2.72) *independent of the funding leg*, while the no-risk-leg controls are flat (USDJPY $t=1.02$, CHFJPY 1.37, EURUSD 1.44) — the controls failing is the proof it's real, not data-mining. It's NOT a JPY thing (JPY is just a clean funding leg; Nikkei works as high-beta risk, not as "Japanese"). **Broadened the live retail_fear FX basket to AUDJPY + NZDJPY + AUDUSD** (`signals.py` + MT5 EA).
- **2026-06-03 — Deletion short DIED at full coverage; AUDJPY re-confirmed; sweep de-crypto'd.** (1) Targeted-profiled all 487 equity pump-authors (25%→100% relevant coverage). The deletion short **flipped negative** — gone ≥ 0.20 → -19.3%/10d ($n=87$), young_frac short -39%→-87% — dominated by squeeze tails (IXHL -625%) the 25% sample had missed. The +14.3% was a **coverage-bias artifact**. Dead as a stock short; puts-only. `distribution_short` (price-rolling-over filter) is the survivor. (2) **AUDJPY capitulation re-validated on 2016-26**: +0.48%/10d VIX-gated, **t(exc)=2.93**, +ve 9/10 years (2018 the lone falling-knife miss). Now the most significant edge in the system; unaffected by author work (pure macro). (3) `get_authors_needing_age` now skips crypto-only authors + front-loads ticker-mentioners.
- **2026-06-03 — Crypto leading-indicator test: COINCIDENT, not leading.** Price lead-lag BTC↔SPY ≈ 0 (same-day +0.17 but $k \neq 0$ noise), weekend test $t=1.2$, crypto sentiment coincident with equity RRAI (peaks $k=0$). Combined crypto+equity RRAI does NOT improve capitulation — *dilutes* the AUDJPY edge (+0.55%→+0.45%). Crypto's value = its own pump/meme universe, not macro timing. (See "Crypto — coincident" section.)
- **2026-06-03 — RRAI extended to 2016-06 (525 caps, +190) → fear-gate STRENGTHENED.** Backfill landed (aggregate_daily 2016-06→2026, 3,653 days; +2018 Volmageddon/Q4 episodes). Re-ran `backtest.fear_gate`: fear-only excess grew SPY +0.32→+0.57, QQQ +0.41→+0.80 (effect grew OOS = real-edge signature); episode-level now **significant on BOTH legs and they agree** (SPY $t=2.77$, QQQ $t=2.37$, $k=63$) — the prior SPY $\neq$ QQQ noise-floor flag is resolved. Day-level still marginal (1.6-1.9). The LIVE `trend+fear` gate is still ≈ 0 (SPY +0.10/QQQ +0.11) ⇒ **drop the trend filter, use fear-only**. AUDJPY confirmed best ungated.

- **2026-06-02 — Power analysis + RRAI history extension (in progress).** Quantified the fear-gate's data need: the 94 fear-days are ~40 independent episodes; day-level it's $t \approx 0.6$ (needs $\sim 9\times$ more obs for $p < 0.05$, $\sim 19\times$ for power — impractical), but episode-level it's $t = 2.28$ (SPY)/1.85 (QQQ) — significant-but-fragile (COVID-cluster-driven; $SPY \neq QQQ \Rightarrow$ noise floor). Cheapest lever = more *independent fear episodes* via **backward** history (Wikipedia fear data already reaches 2015). Extended study prices to max and **kicked off the RRAI aggregate backfill 2016-06→2020** (wallstreetbets/stocks/pennystocks) to add the 2018 Volmageddon + Q4-2018 episodes; will re-run `backtest.fear_gate` on the longer series when it lands.
- **2026-06-02 — Committed EXCESS study settles the fear-gate (`backtest/fear_gate.py`).** The proper frame (mean excess vs buy-&-hold, the +0.36pp's own metric). Three findings: (1) the +0.36pp **reproduces** as a point estimate (SPY +0.32 / QQQ +0.41 fear-only excess) — not fabricated; (2) it's **not significant** (excess $t \approx 0.6$, $n \approx 94$); (3) the LIVE **trend+fear** combo is *worse* — dilutes SPY ($\rightarrow +0.09$), flips QQQ negative ($\rightarrow -0.20$) — because the trend filter strips the downtrend bounces that carry the excess. Reconciles the MT5 "loss": MT5 sums absolute P&L (COVID-cluster dominates), excess means-averages — so the earlier "drag" was a frame artifact. Gate stays ; fear-only > trend+fear for excess; AUDJPY stays ungated (the only significant edge). Exporter refactored to share the study's `capitulation_sets()` (MT5 + Python test identical dates).
- **2026-06-02 — MT5 backtest of the fear-gate → it's a DRAG, not a rescue.** Extended `--export-signals` to emit 4 point-in-time capitulation date-sets (baseline / `fear_z ≥ 0.5` / 200-DMA-trend / trend+fear) and ran each through the MT5 Strategy Tester on US500 (2020-26) via `RedditMacro_US500_*.ini`. Adding `fear_z` made absolute returns *worse* (alone $-\$1045$, with trend $-\$372$ vs trend-only $+\$21$) — fear spikes fire long into tops (Feb-2020). Only the trend filter helps (DD 13.6→7.1%). Caveat: MT5 = absolute long-only, the +0.36pp was excess — so not a direct refutation, but the gate is downgraded to  doubtful and needs a committed *excess-based* study to settle (maybe drop `fear_z`, keep trend).
- **2026-06-02 — Doc-completeness pass.** Audited the doc against the live registry + all backtest modules. Found one missing edge — the **Wikipedia fear-confirmation gate** (live: gates the equity legs of `retail_fear`, `fear_z ≥ 0.5`, +0.36pp rescue) — now documented, *honestly flagged as not yet committed-backtested*. Also added the full live-screen list (`fresh_pump_alert`, `coordination_flag`, ...) to the scoreboard and recorded the broad "attention-leads-price" long as a failed up-leg test.
- **2026-06-02 — New PIT-tradeable short found: `distribution_short` (`backtest/microstructure.py`).** Built the missing **new-author recruitment** feature and tested demand-quality timing signals. The decisive lesson: microstructure timing (recruitment-exhaustion / deceleration / divergence) is **worthless on the broad attention universe** (it drifts flat-to-up, costs dominate) but **powerful inside the concentrated/manufactured subset** — exactly where the deletion edge lives. Winner = concentrated pump + euphoric sentiment + price already rolling over $\rightarrow$ **+7.2%/10d net, $t = 2.70$ (date-clustered $t = 2.52$, 47 indep. dates), dose-response, survives $2\times$ costs**; control loses -4.1% . Unlike the deletion gate it uses **no hindsight** $\rightarrow$ tradeable. Recorded negatives (broad timing, recruitment-alone, up-leg ride) to avoid re-runs.
- **2026-06-02 — Deletion-gated short forensically validated.** Author coverage 8%→20% (2,973 gone). Re-backtest: `gone_frac ≥ 0.20` $\rightarrow$ +14.3%/10d short (83% win, $t = 2.57$, +ve all regimes); the *control* (un-gated concentrated pumps) *loses* -10.8% — so the deletion gate flips the trade, confirming the A2 thesis. **Caveat:** `gone_frac` is hindsight (no Reddit deletion timestamp) $\rightarrow$ not yet PIT-tradeable; the `young_frac` PIT proxy is still too sparse ($n = 3-7$). Also shipped `scraper/finra.py` (FINRA Reg SHO daily short-volume, 475k rows) + z-score `squeeze_screen()` — learned the *absolute* short-ratio is market-maker noise (median 0.49), so the screen uses per-ticker z-score anomaly + chatter.
- **2026-06-01 — Instrument scan (17 MT5 instruments).** `retail_fear` generalises across the risk-on basket; commodities express it best (Silver +2.2pp lumpy, Gold +0.6pp robust), AUDJPY steadiest. Oil/crypto dropped. Basket = AUDJPY + Gold + indices + small Silver; EA + feed updated.
- **2026-06-01 — Macro R&D + MT5 framework.** Benchmarked candidates vs buy-and-hold: only the `retail_fear` contrarian family beats it (best AUDJPY +0.47pp/10d excess); momentum/froth = no edge, all shorts fail. Built `SignalClient.mqh` + `RedditMacro_EA.mq5` (strategy-selectable), multi-strategy `/signals` feed; retired `RedditRRAI_EA.mq5`.
- **2026-06-01 — Initial document.** Phase 0 (feature store + friction-aware backtester) shipped. RRAI capitulation-long validated (best via AUDJPY, VIX-gated); pump-ride/naive-dump-short/organic-long failed; fade-organic marginal; deletion-gated short awaiting author coverage (8%). `/signals` + MT5 EA live.